

Rectifiers

PE Converter Categories

- AC-DC: Rectifier
- DC-DC: Chopper
- AC-AC Voltage Controllers
- Cycloconverters
- DC-AC: Inverter

Rectifiers

Classification (for both single phase and three phase full wave rectifier)

- Uncontrolled (all diodes)
 - Half Controlled (half diode+half thyristor)
 - Fully Controlled (all thyristors- **HOW?**)
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- Important: Difference between half wave rectifier and half controlled rectifier.

Single Phase Controlled Half Wave Rectifier (One Pulse Converter)

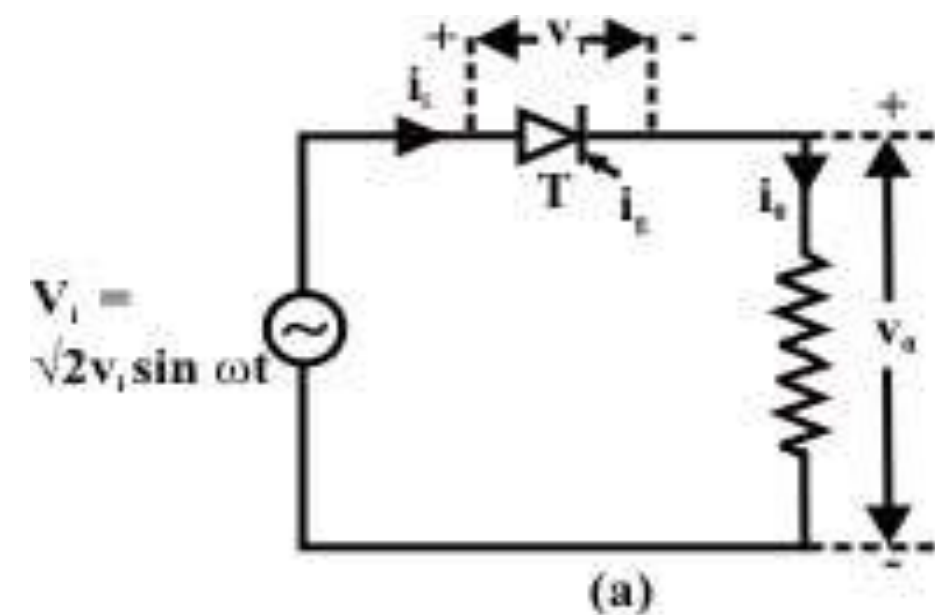
Thumb Rule for ease in waveform drawing

Switch State	Passive Load		Active Load	
	V_{Sw}	V_{Load}	V_{Sw}	V_{Load}
OFF	V_{in}	0	$V_{in}-E$	E
ON	0	V_{in}	0	V_{in}

Loads

- R
- L
- RL
- RL with FWD
- RE
- RLE

- Why not C?



(a) Circuit diagram
(b) Waveforms

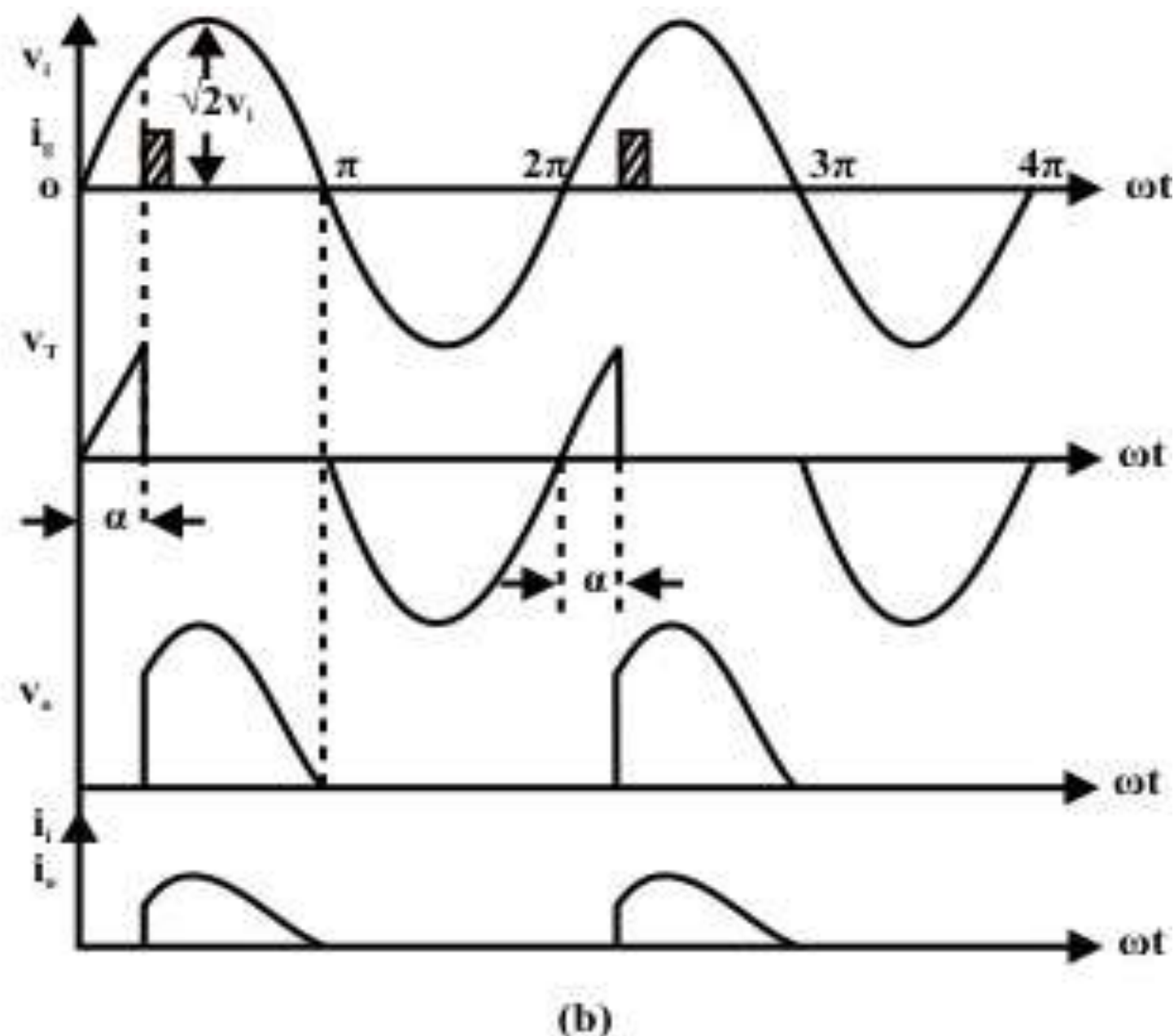
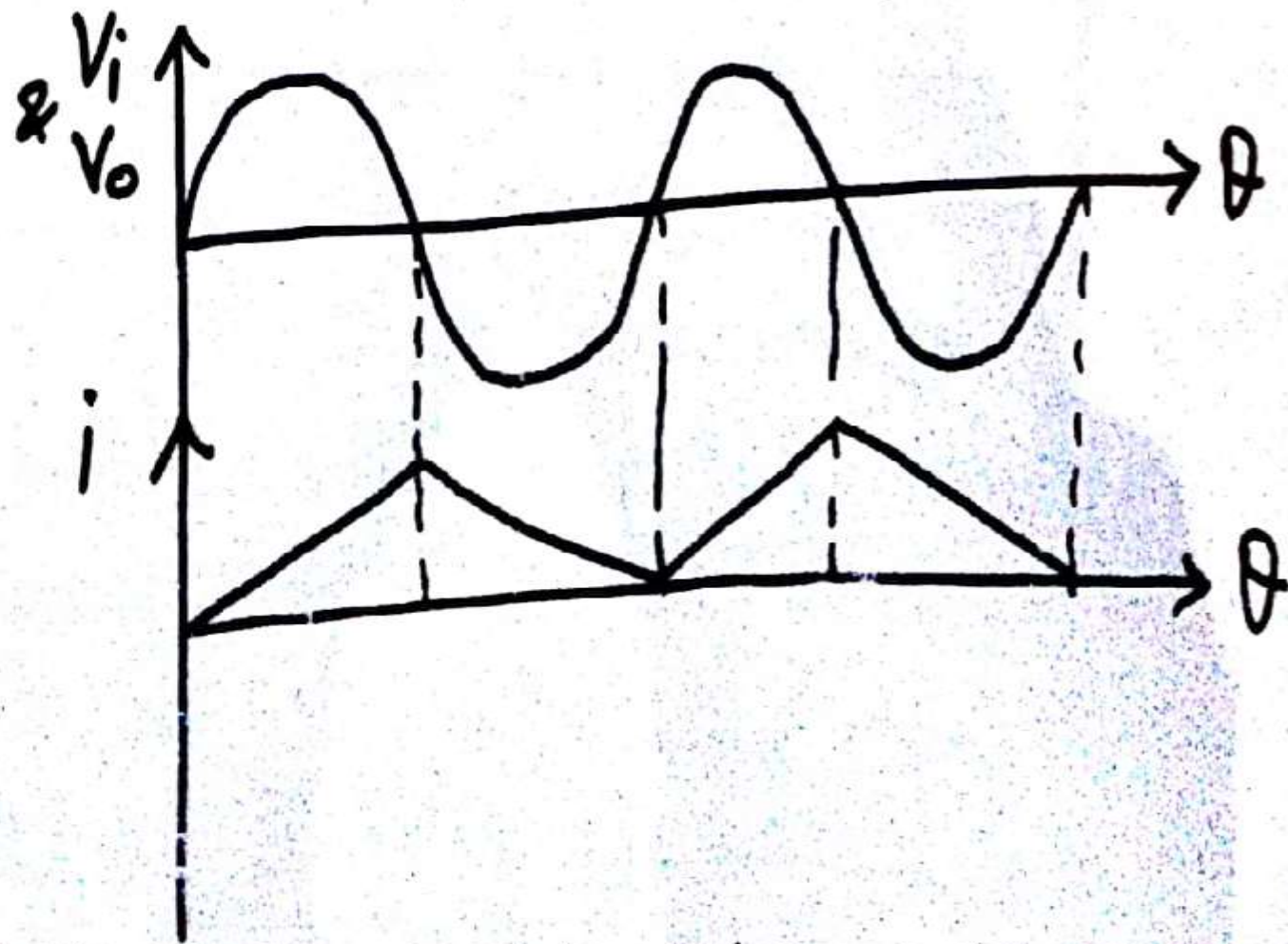
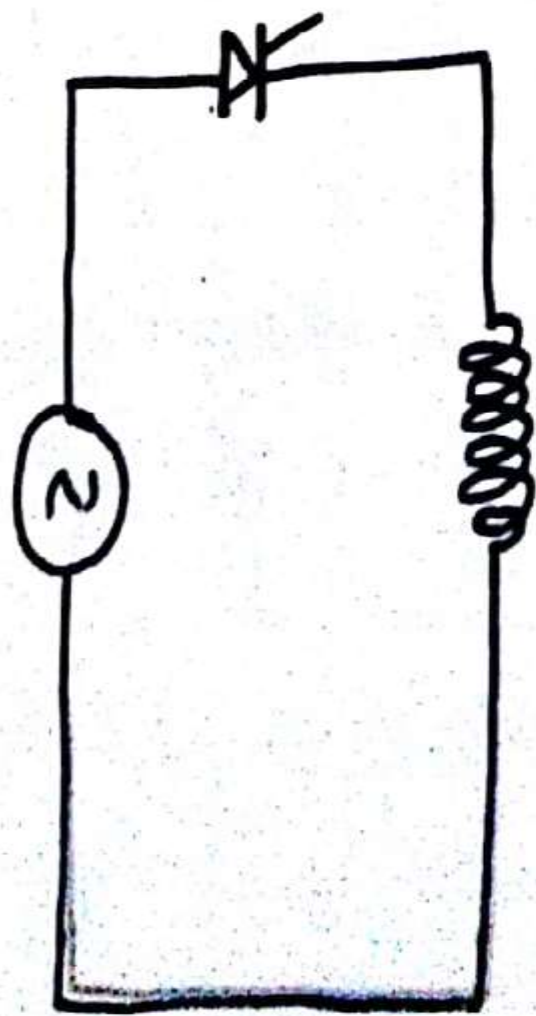


Fig. 10.1: Single phase fully controlled half wave rectifier supplying a resistive load

SINGLE PHASE HALF WAVE CIRCUIT WITH L LOAD



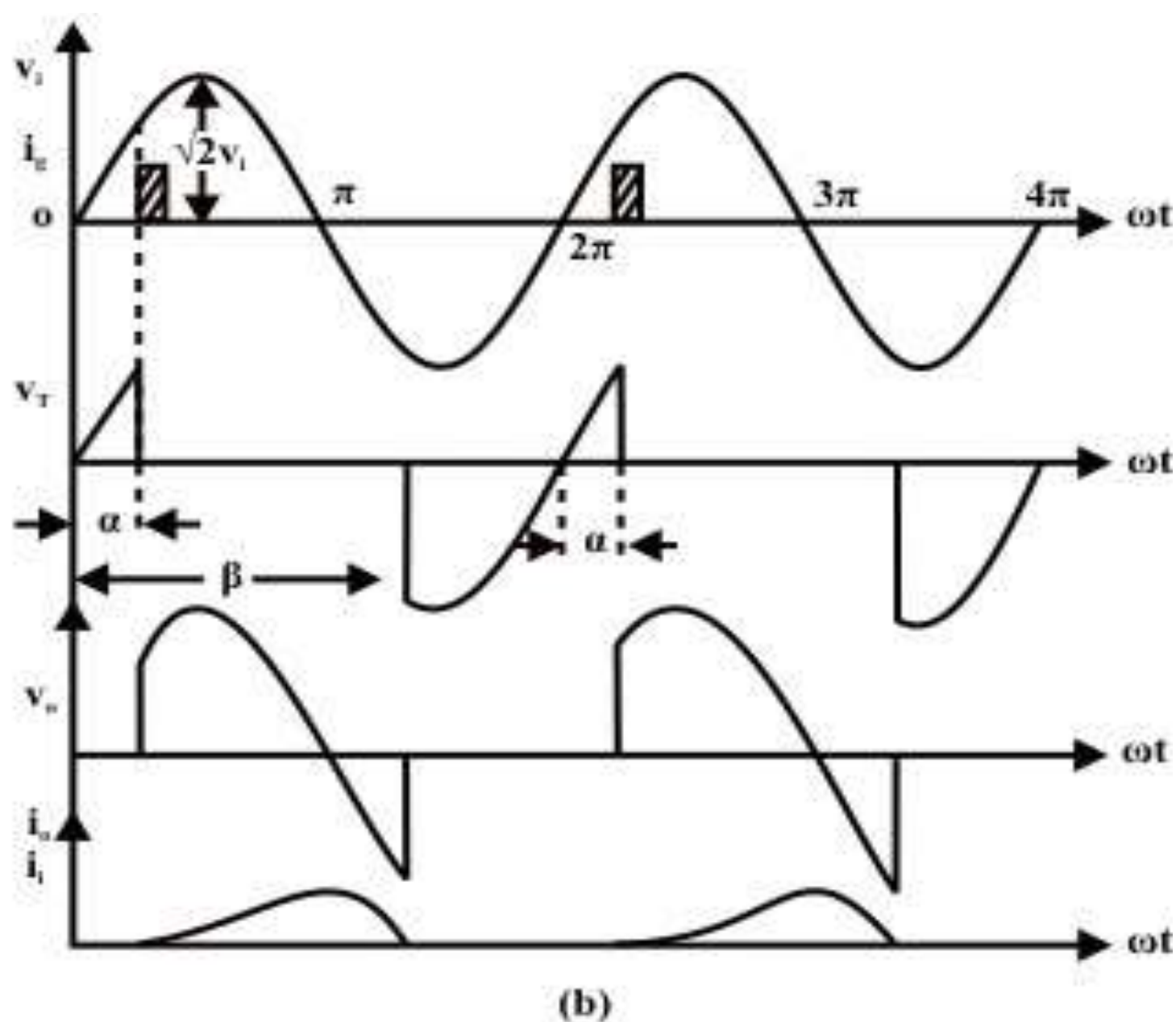
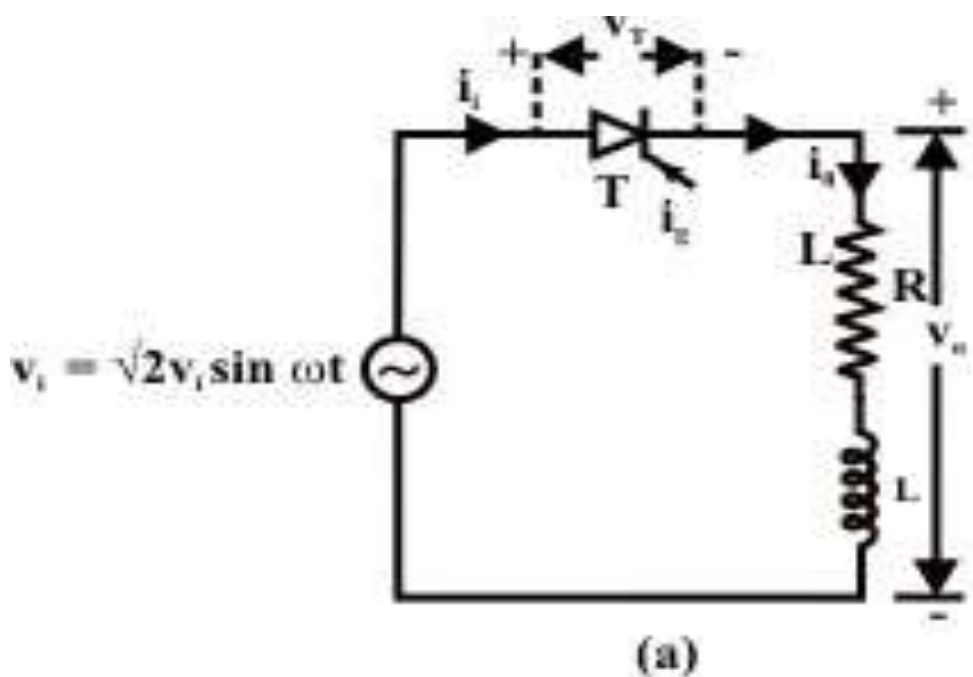
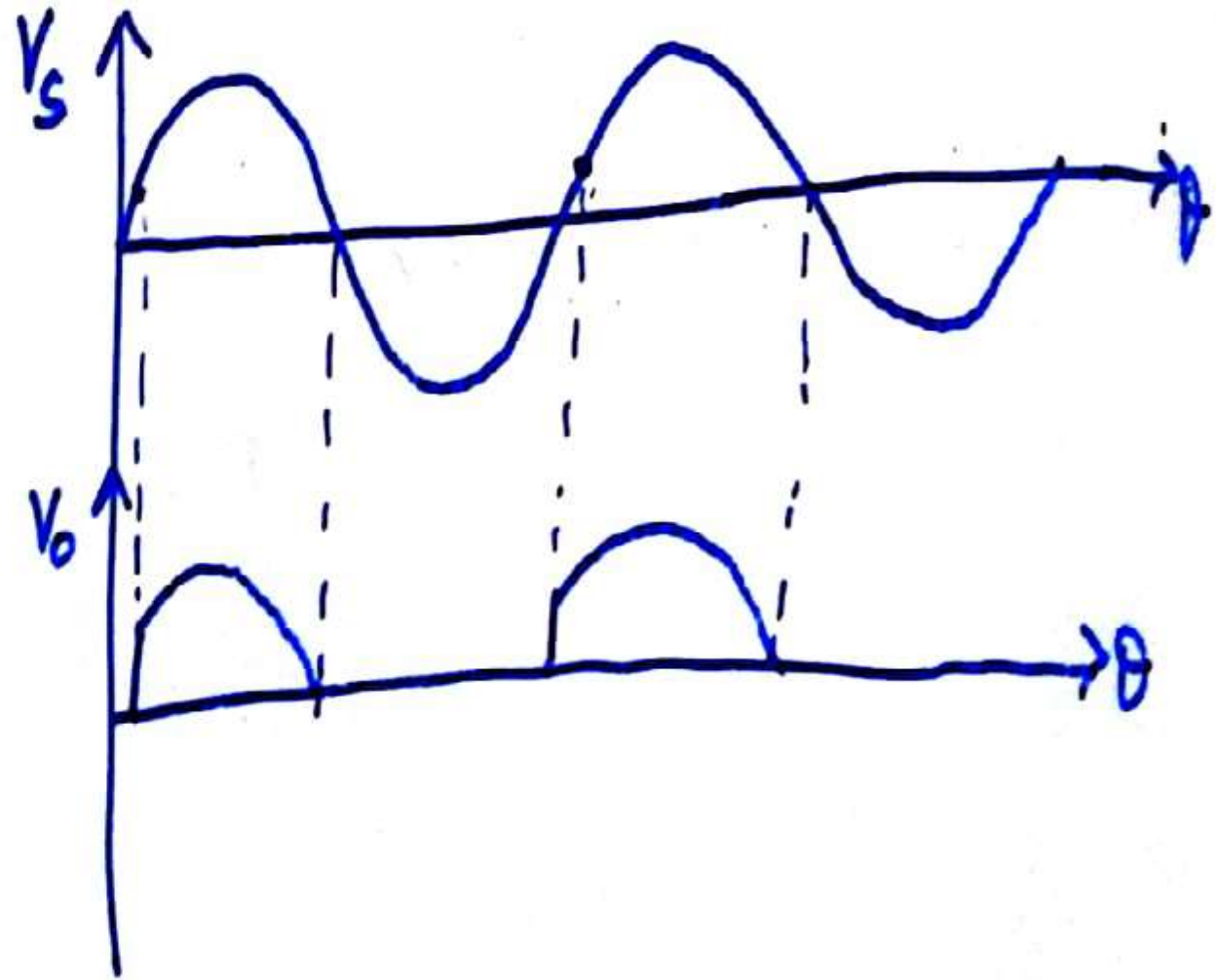
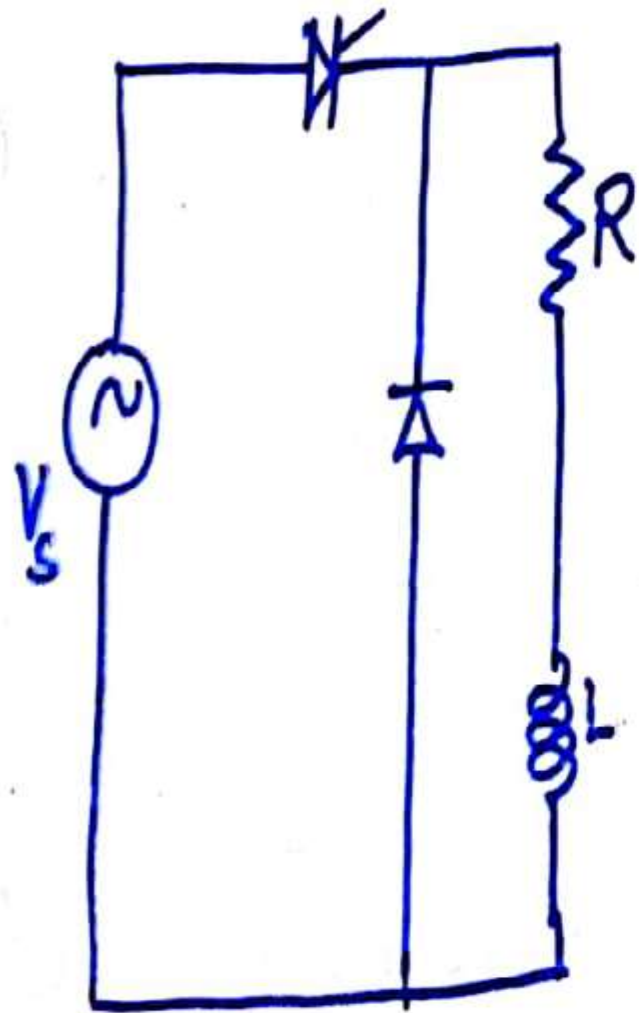
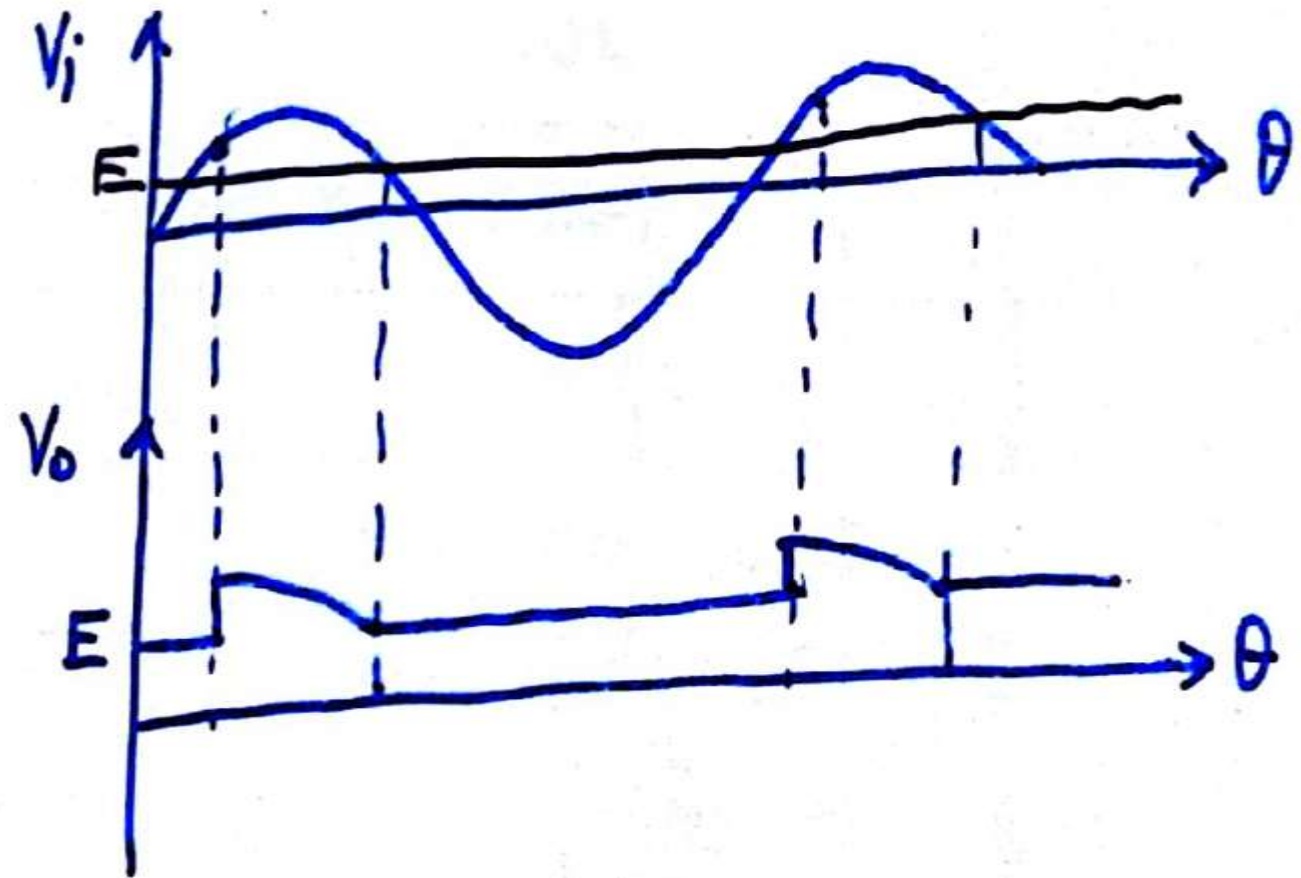
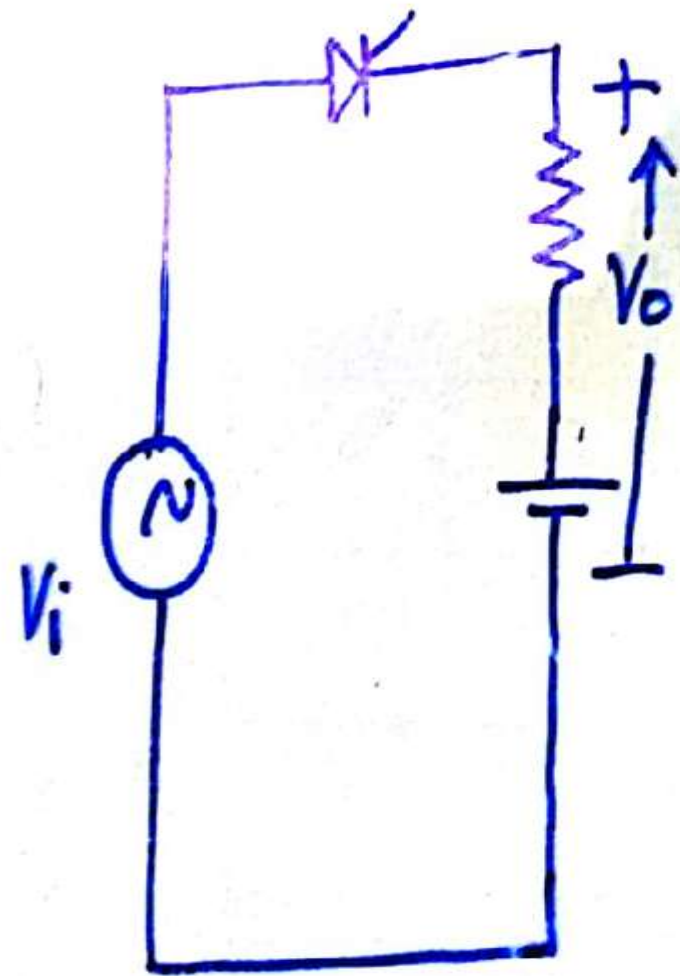


Fig. 10.2: Single phase fully controlled half wave rectifier supplying a resistive inductive load

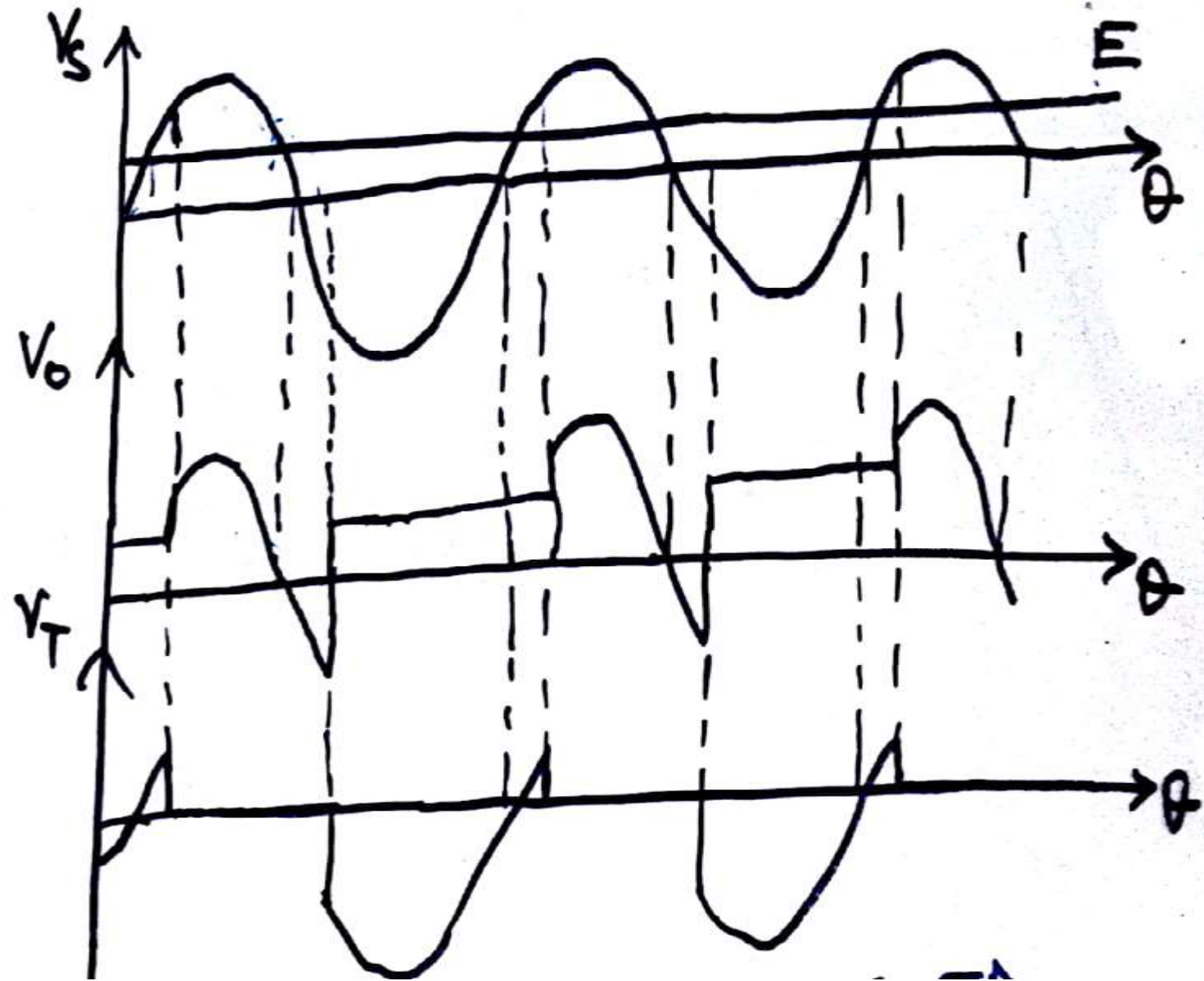
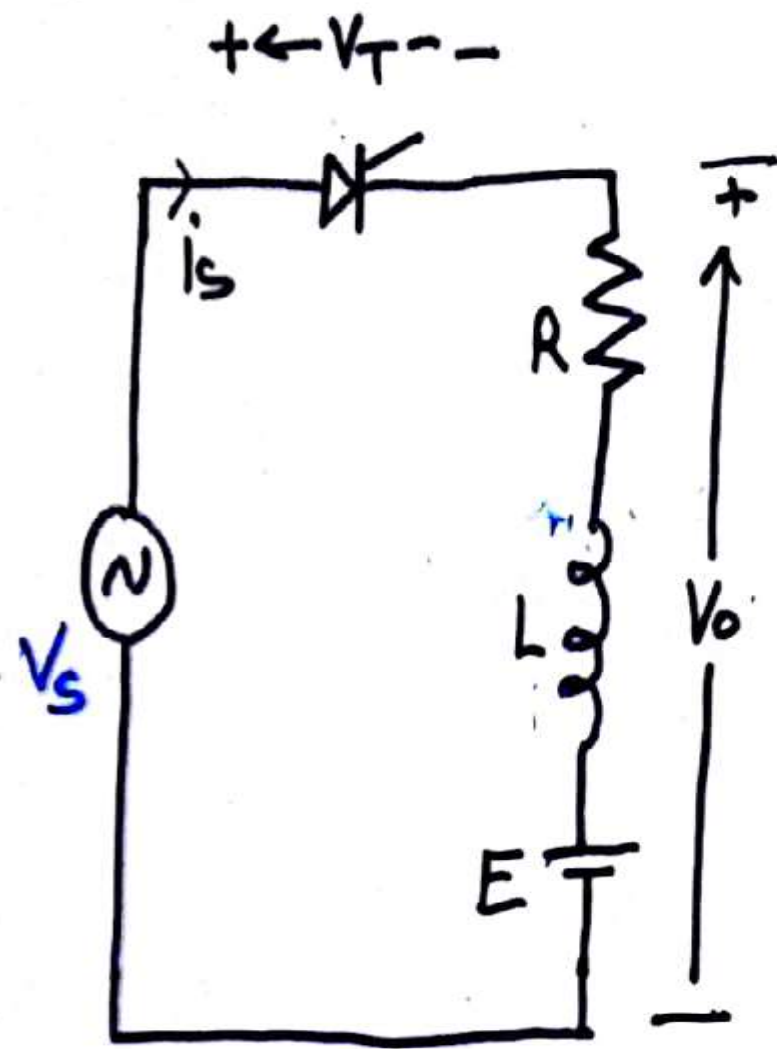
SINGLE PHASE HALF WAVE CIRCUIT WITH RL LOAD & FD



SINGLE PHASE HALF WAVE CIRCUIT WITH RE LOAD



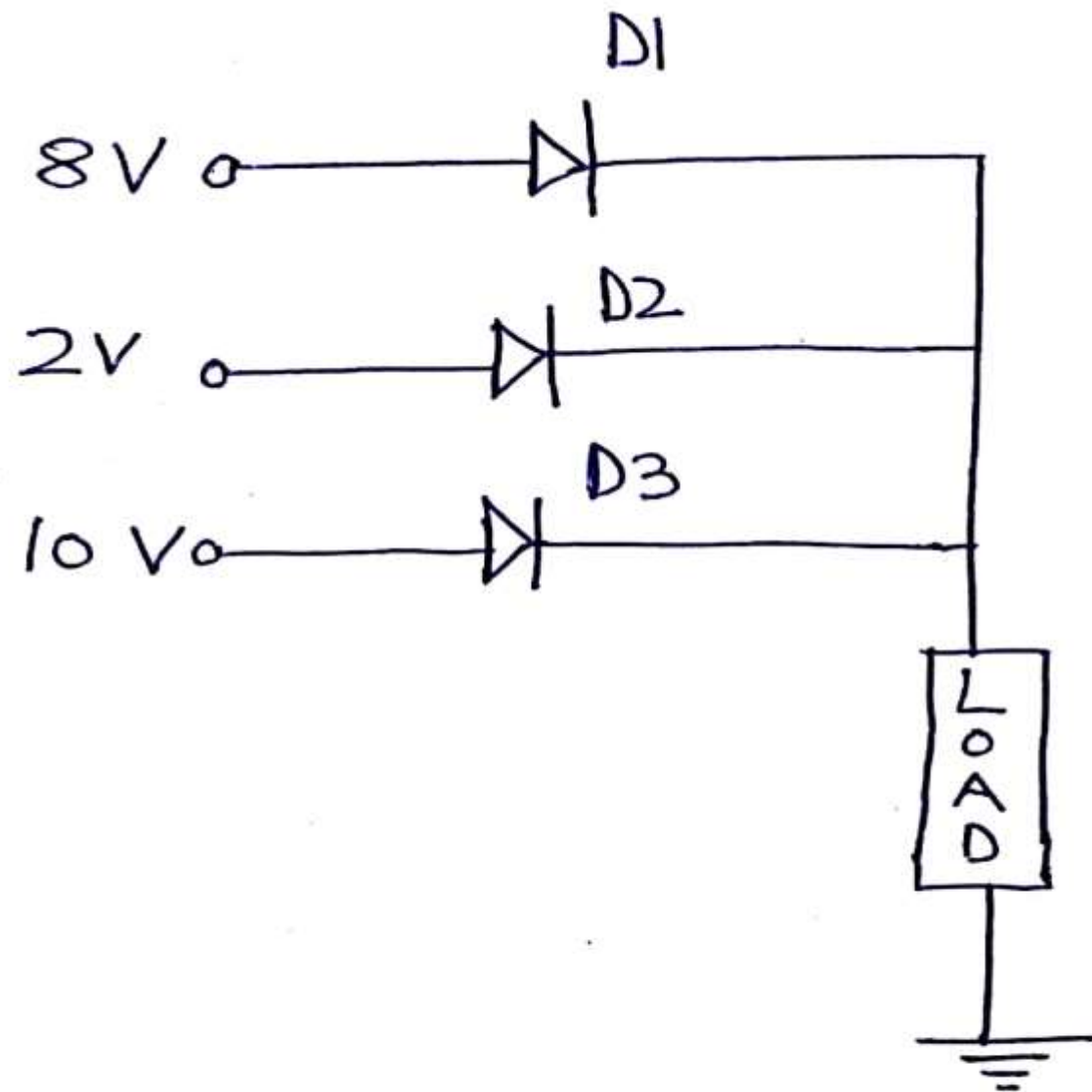
SINGLE PHASE HALF WAVE CIRCUIT WITH RLE LOAD (43)



Single Phase Fully Controlled Full Wave Rectifier (Two Pulse Converter)

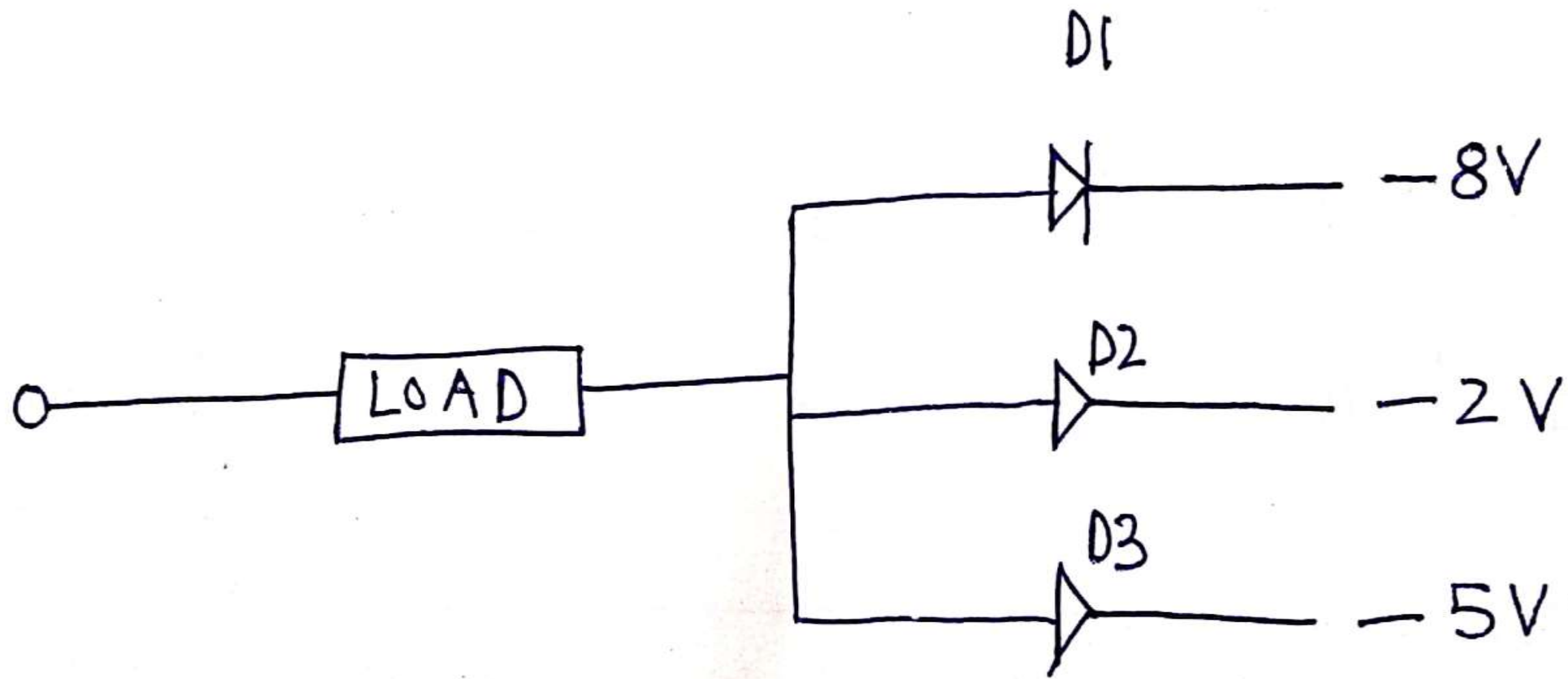
Thumb Rule for ease in waveform drawing

- When T1 and T2 conduct, the load voltage and current are a replica of the source voltage and current.
- When T3 and T4 conduct, the load voltage and current are the negative replica of the source voltage and current.

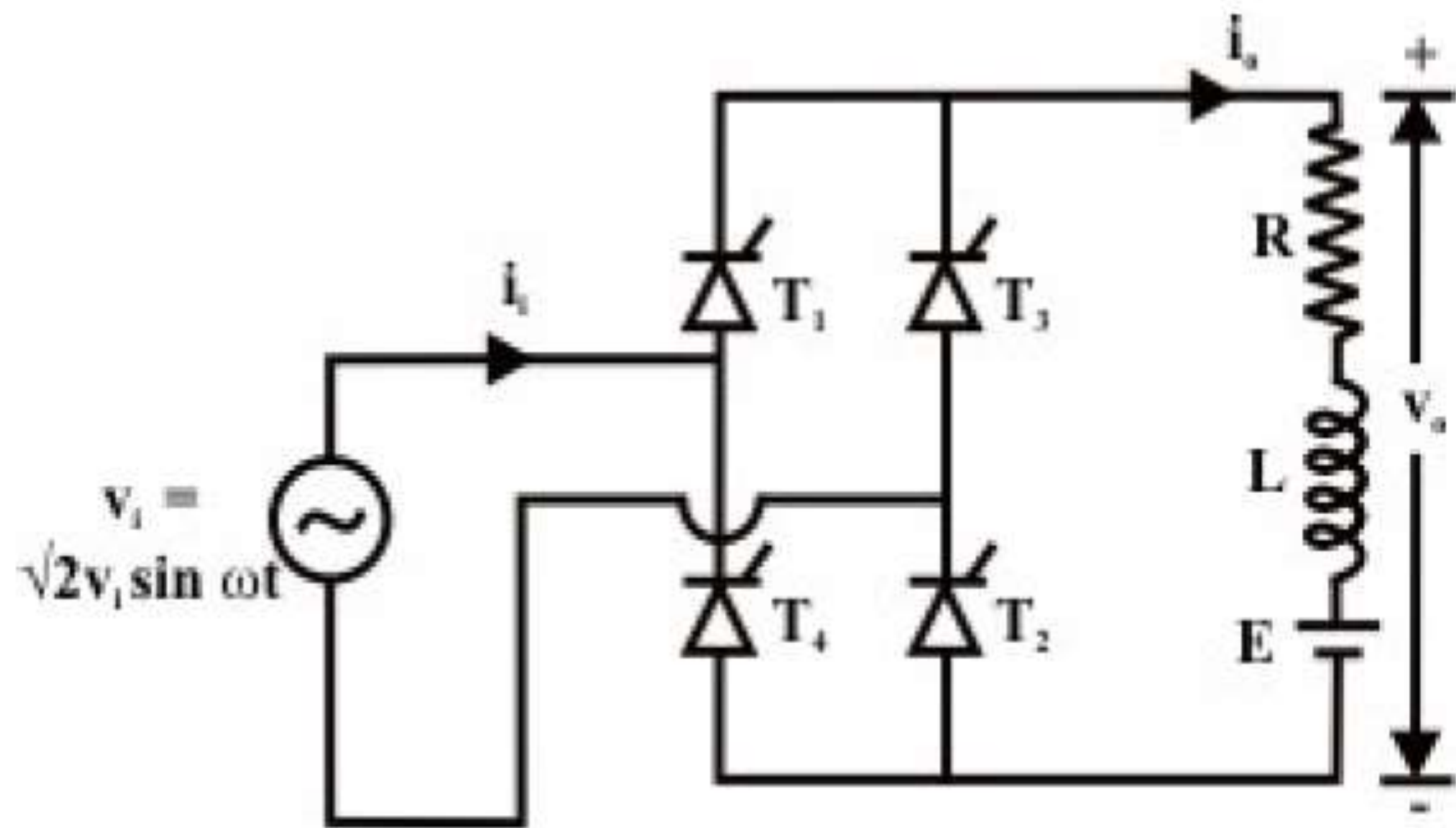


That diode will conduct whose anode potential is the highest.

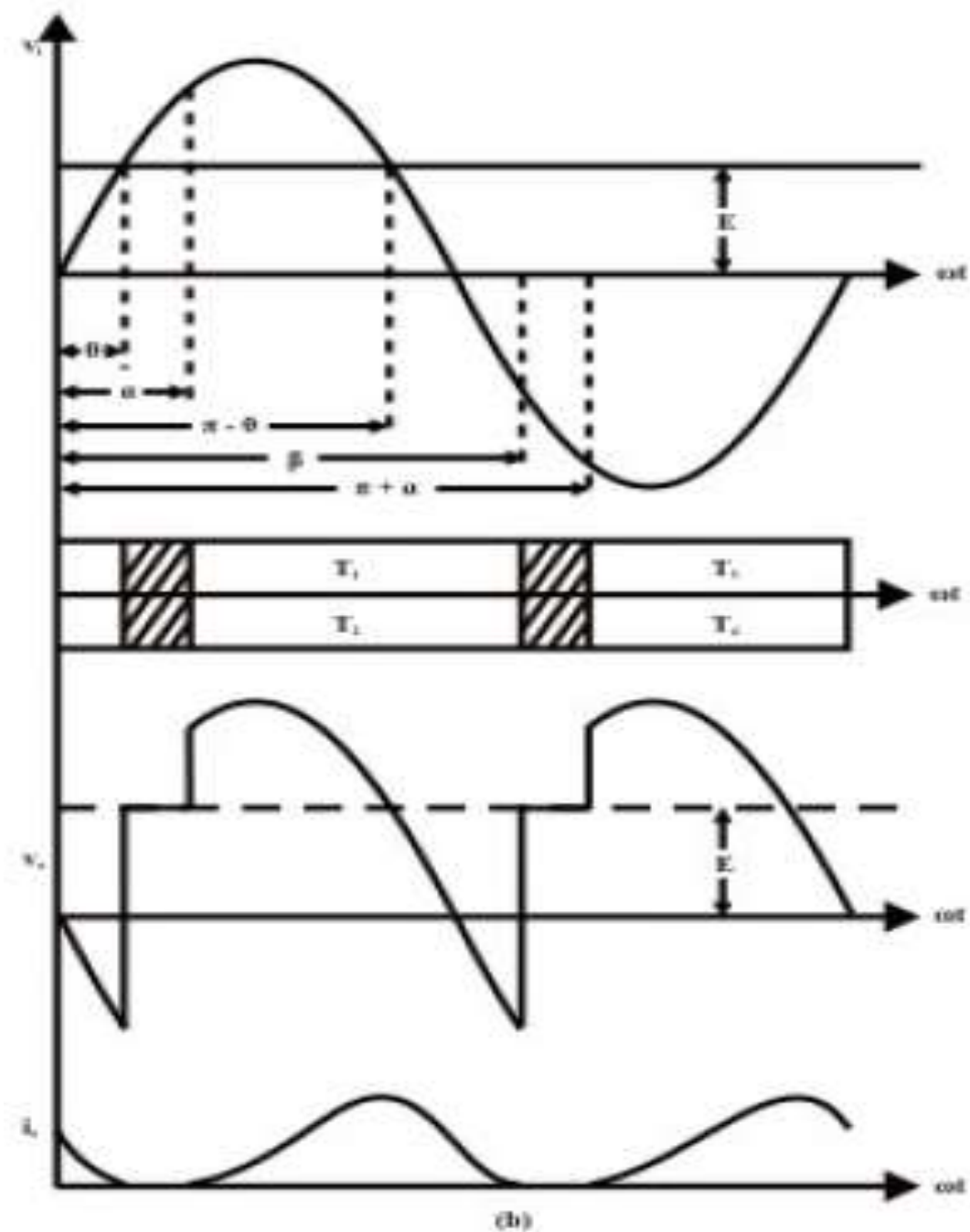
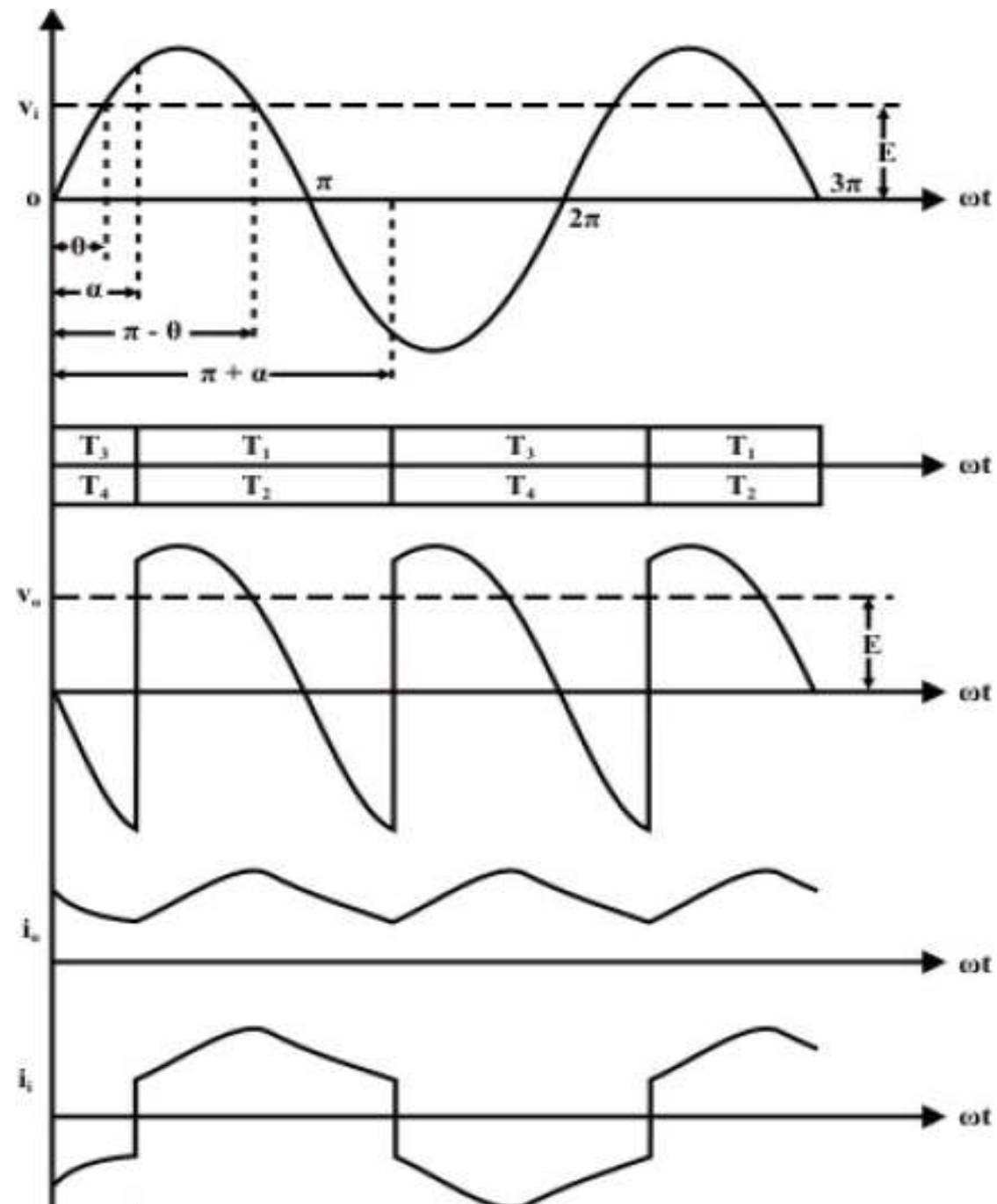
Common Cathode Configuration



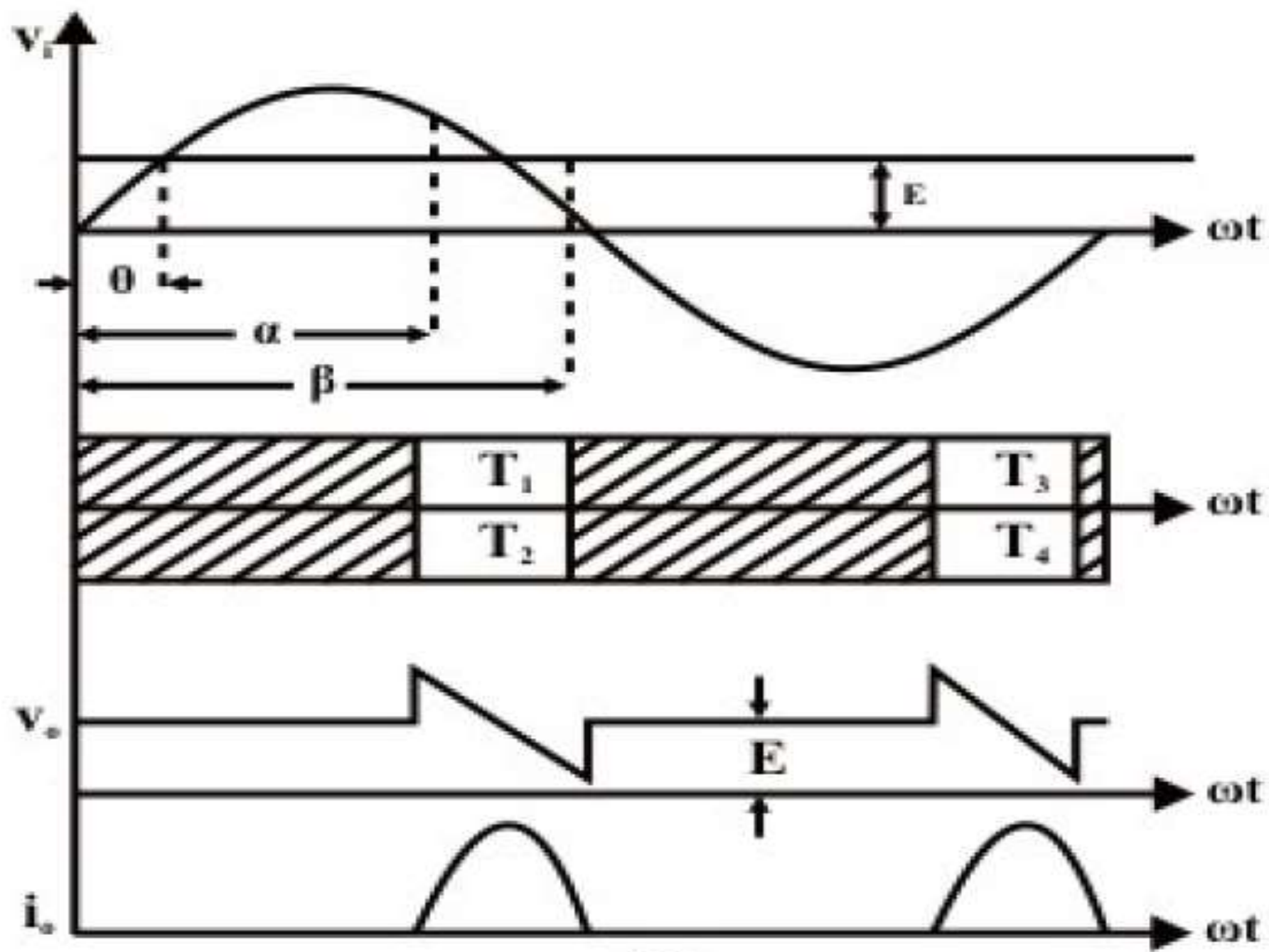
Common Anode Configuration

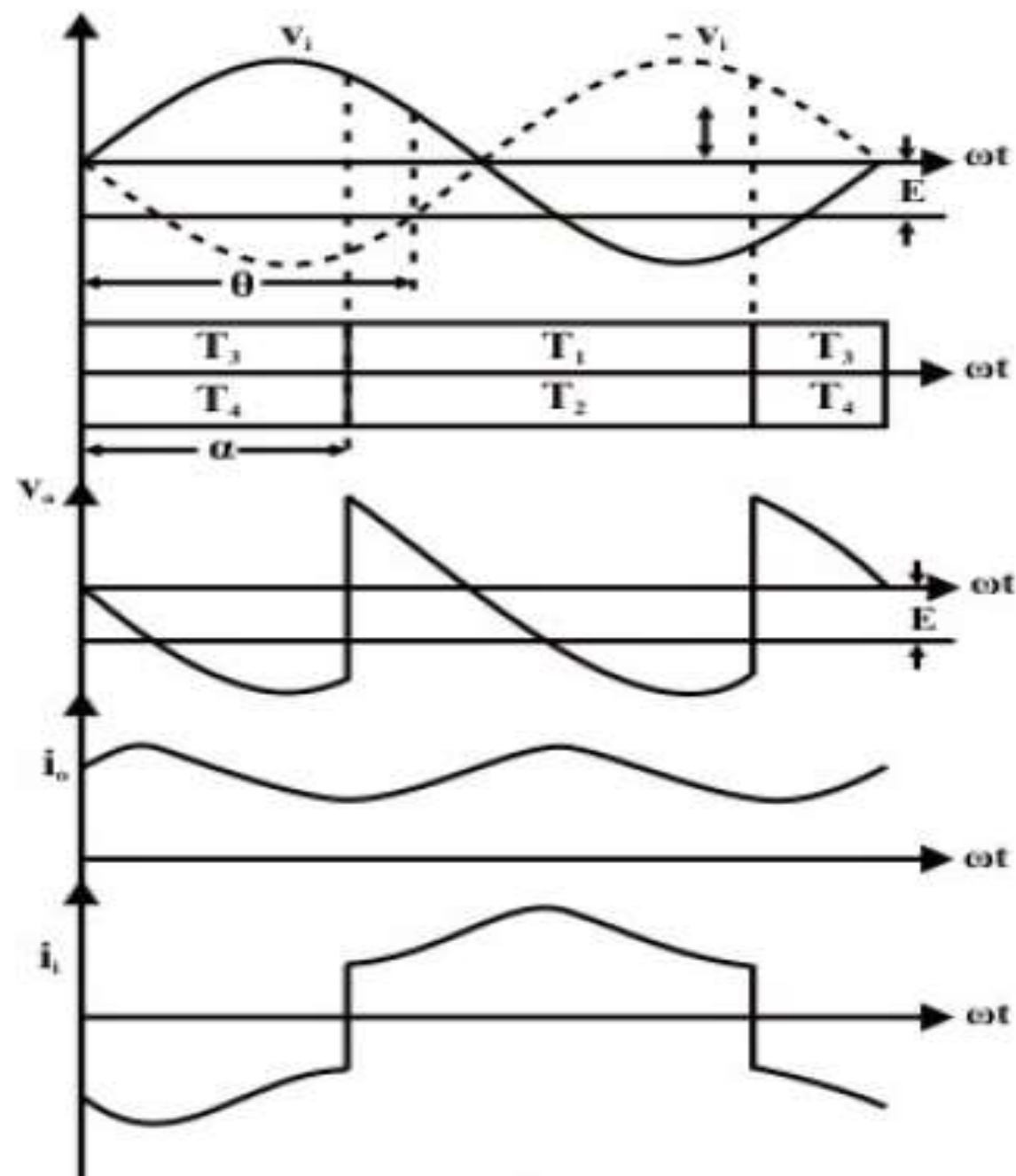
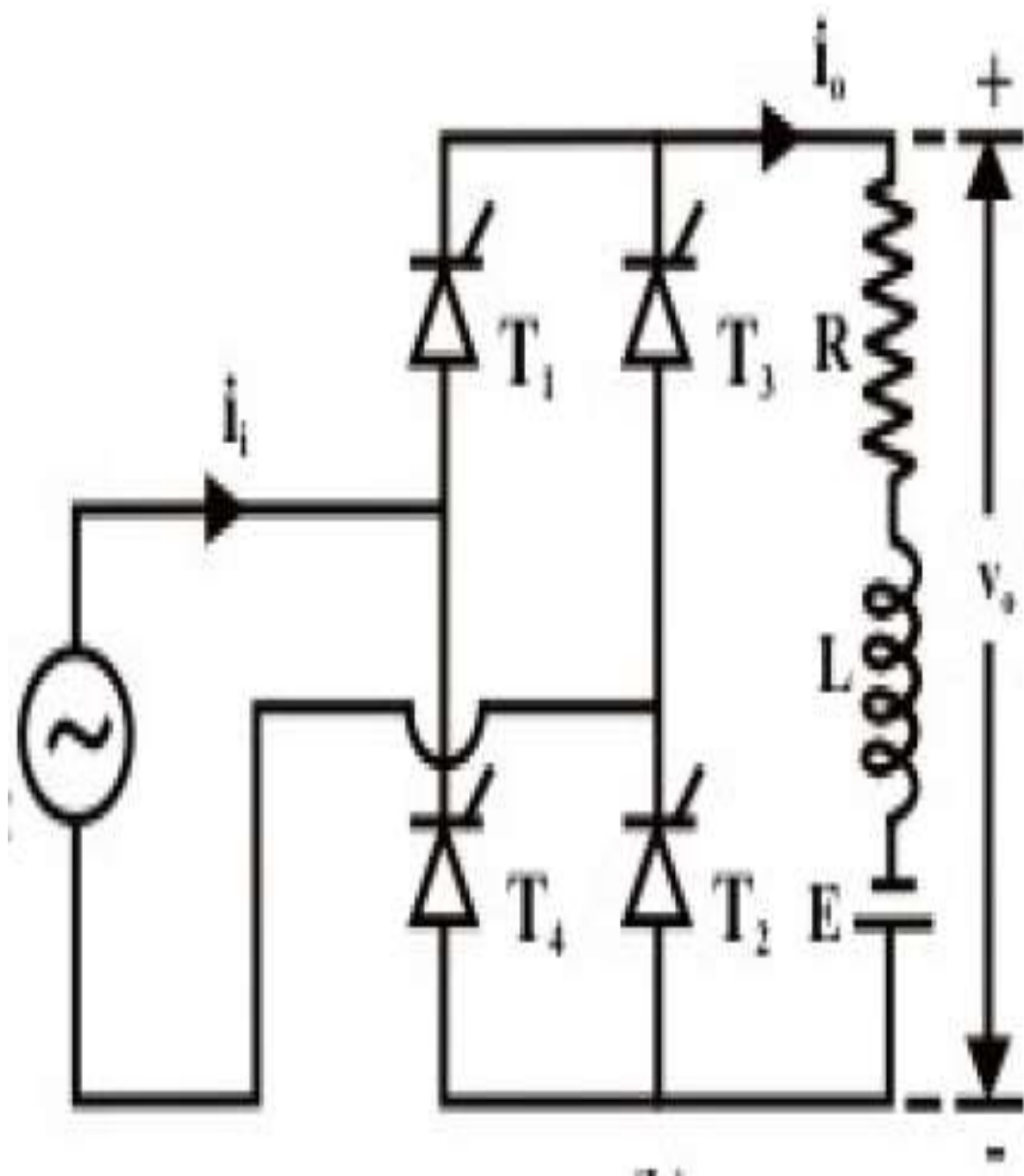


CCM vs DCM

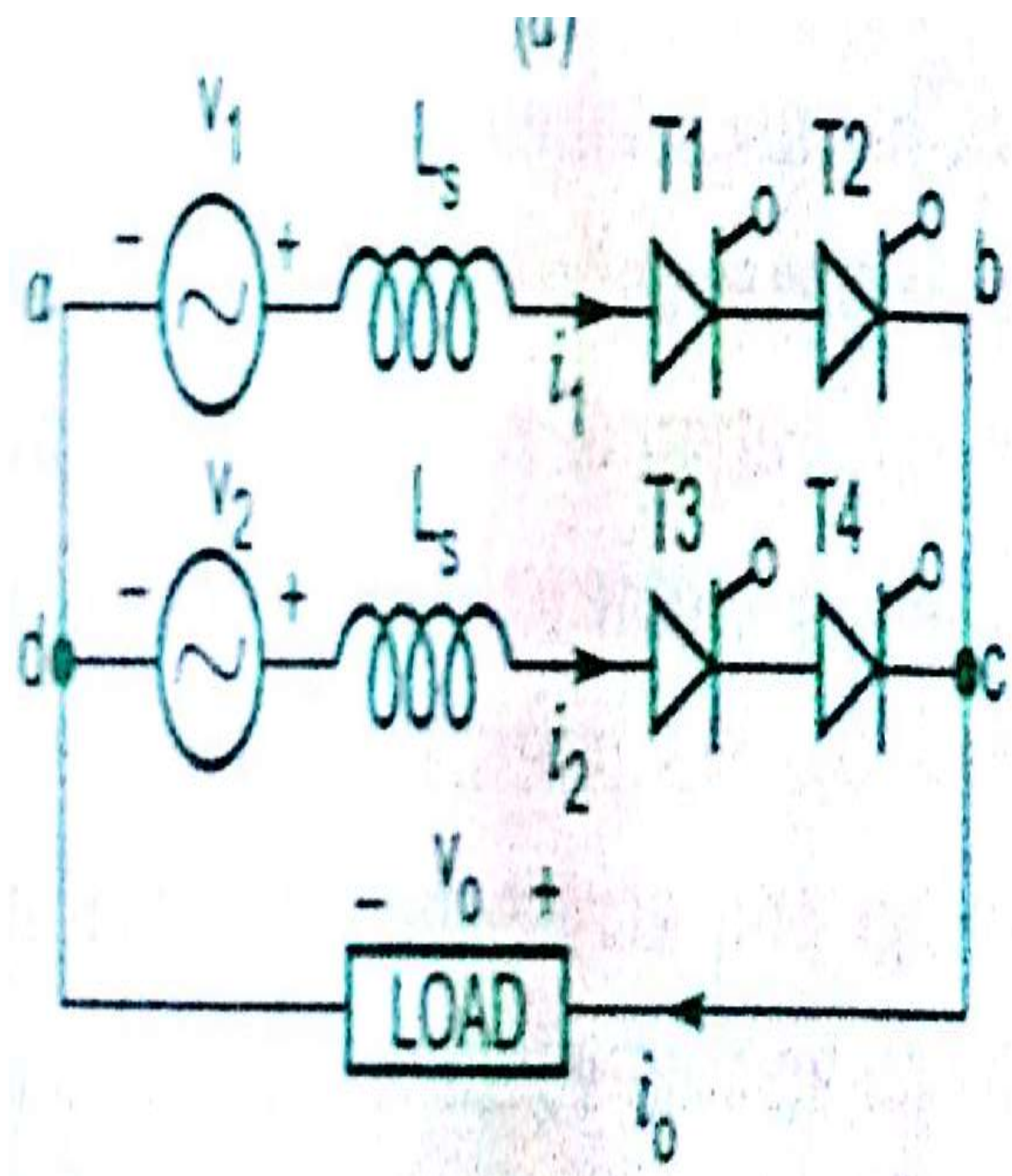
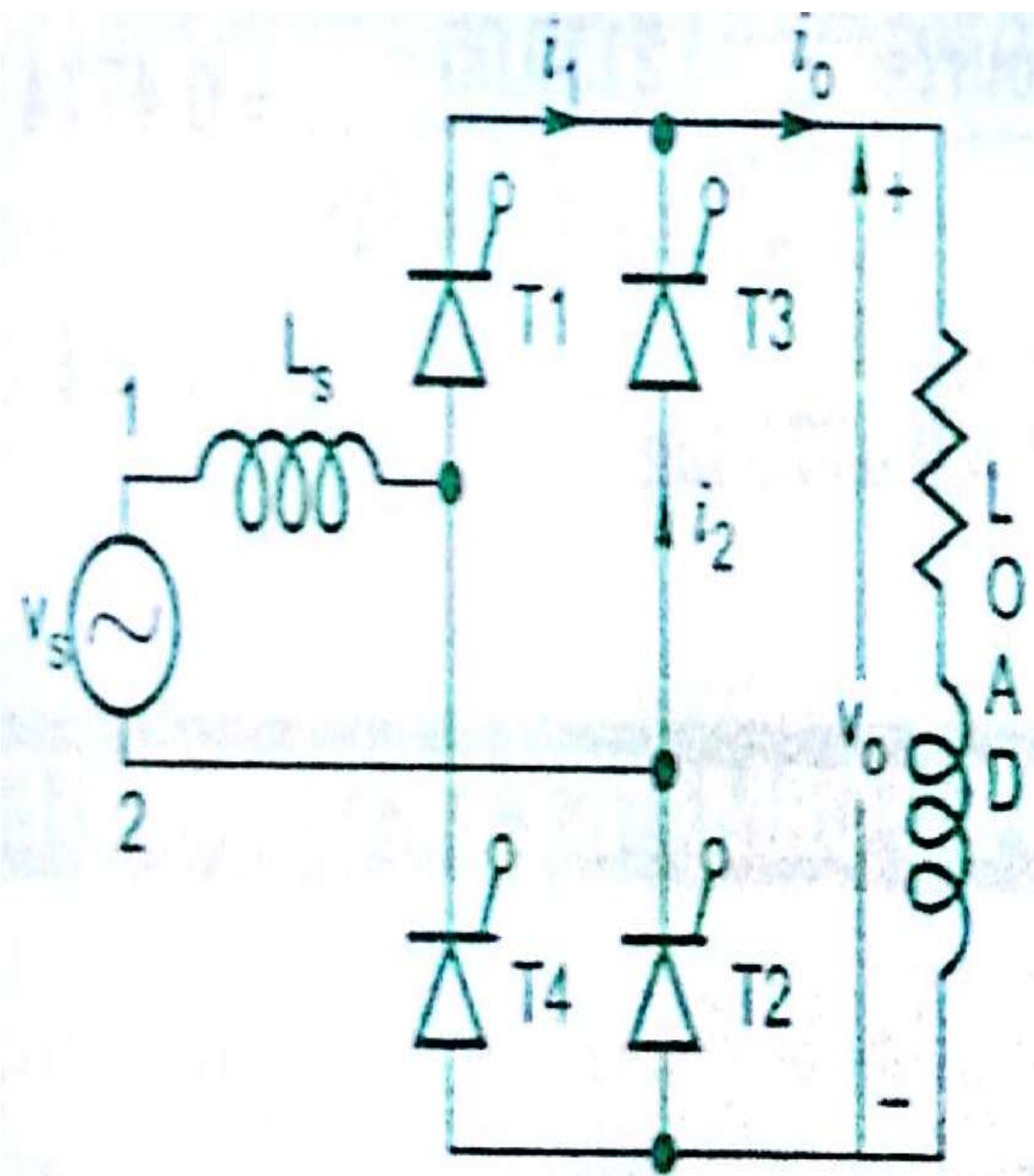


Inverter Mode





Effect of Source Inductance



$$V_1 - L_s \frac{di_1}{dt} = V_0 \dots (i)$$

$$V_1 - L_s \frac{di_1}{dt} + L_s \frac{di_2}{dt} - V_2 = 0$$

$$\Rightarrow V_1 - V_2 = L_s \left(\frac{di_1}{dt} - \frac{di_2}{dt} \right)$$

$$\Rightarrow \frac{di_1}{dt} - \frac{di_2}{dt} = \frac{V_1 - V_2}{L_s} \dots (ii)$$

$$i_1 + i_2 = I_0 \dots (iii)$$

$$\Rightarrow \frac{di_1}{dt} + \frac{di_2}{dt} = 0 \dots (iv)$$

From (ii) & (iv)

$$\frac{di_1}{dt} = \frac{V_1 - V_2}{2Ls}$$

$$\text{or } Ls \frac{di_1}{dt} = \frac{V_1 - V_2}{2} \dots (v)$$

$$V_1 = -V_2 \dots (vi)$$

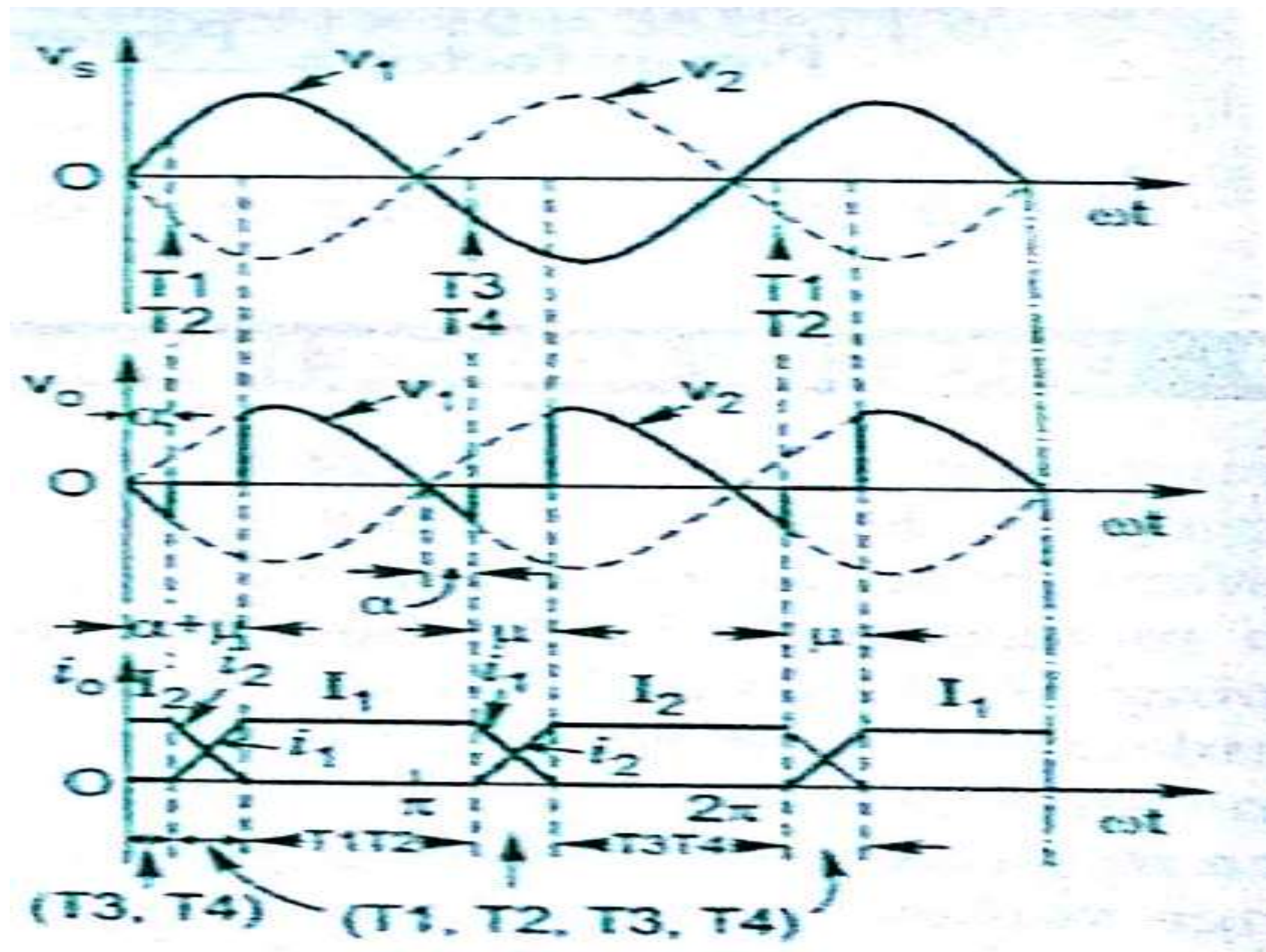
From (v) & (vi)

$$L_s \frac{di_1}{dt} = V_1 \dots (vii)$$

Substituting (vii) in (i) we get

$$V_1 - V_1 = V_0$$

$$\Rightarrow \boxed{V_0 = 0}$$



Single Phase Half Controlled Full Wave Rectifier (Semi Converter)

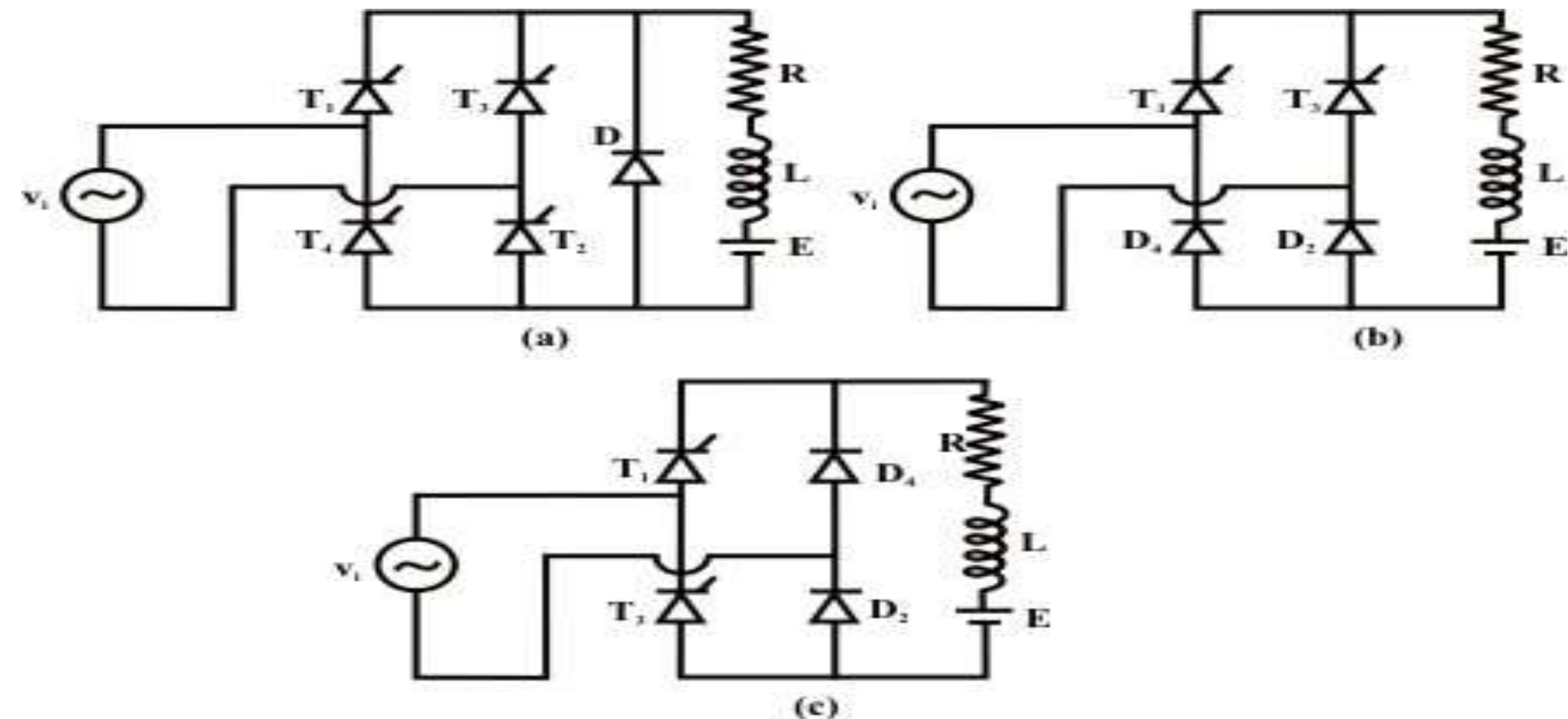


Fig. 11.1: Alternative circuits of single phase half controlled converter.
 (a) fully controlled converter with load side free wheeling diode
 (b) half controlled converter - alternative -1
 (c) half controlled converter - alternative -2.

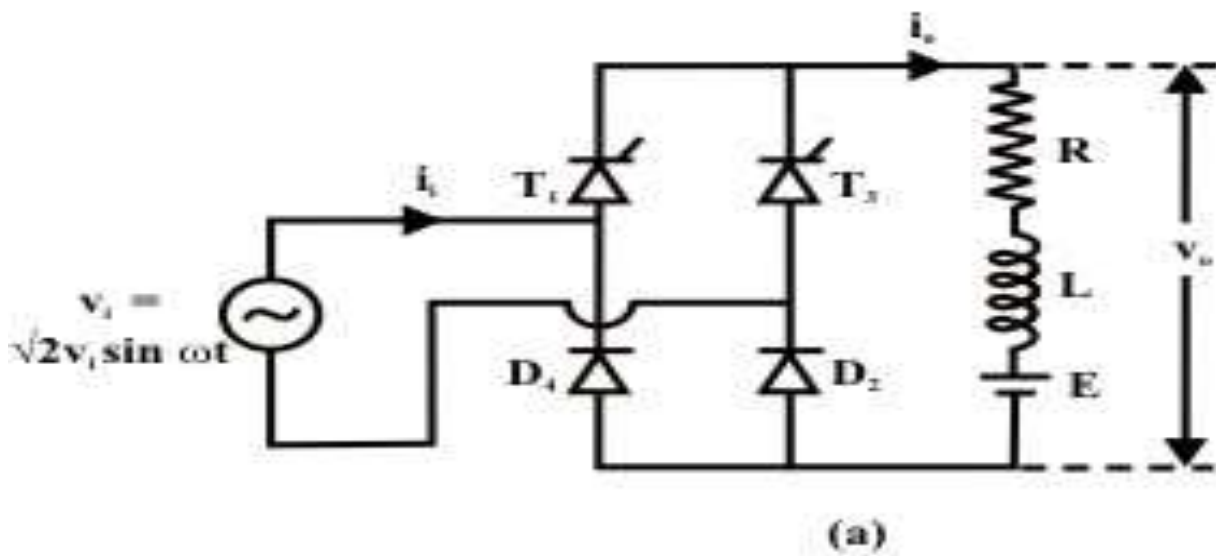


Fig. 11.3: Single phase half controlled converter in the continuous conduction mode
 (a) Circuit diagram
 (b) Waveforms

